

Fig .12.7 (a) DEWATS system at Auroville



(b) RZWT system at Aravind Eye Hospital

inactivates chlorine-resistant microorganisms like Cryptosporidium and Giardia.

Case Study: Auroville, located in South India near Puducherry has been experimenting with natural wastewater recycling systems (Decentralized Waste Water Treatment System (DEWATS)) (**Fig.12.7a**). Such treatment plants have now also been implemented in Aravind Eye Hospital, Puducherry (Root Zone Wastewater Treatment (RZWT)) **(Fig.12.7 b)** and the Chennai Mathematical Institute, Siruseri IT Park, Chennai.

12.8 Organic Farming and Its Implementation

It is a method of farming system which primarily aims at cultivating the land and raising crops in such a way, so as to keep the soil alive and in good health by use of organic wastes (crop, animal and farm wastes, aquatic wastes) and other biological materials along with beneficial microbes (biofertilizers) to release nutrients to crops for increased sustainable production in an eco-friendly pollution free environment.

12.9 Solid Waste Management

Every day, tonnes of solid wastes are disposed off at landfill sites. This waste comes from homes, offices, industries and various other agricultural related activities. These landfill sites produce foul smell if waste is not stored and treated properly. When hazardous wastes like pesticides, batteries containing lead, cadmium, mercury or zinc, cleaning solvents, radioactive materials, e-waste and plastics are mixed up with paper and other scraps and burnt, they produce gases such as dioxins. These gases are toxic and carcinogenic. These pollute the surrounding



G. Nammalvar was a supporter and expert of **organic farming**. He was an agricultural scientist, environmental activist celebrated for his work on spreading **Ecological farming** & **Organic farming**. He was against the use of chemical fertilisers and pesticides. He trained hundreds of farmers in natural farming. Nammalvar was the author of several **Tamil** and **English** books on natural farming, pesticides & fertilisers and was featured in magazines & television programs. He founded the Nammalvar Ecological Foundation for Farm Research and Global Food Security Trust or simply **Vaanagam** at Karur, Tamilnadu. He developed social forest at Ammankurai and the Kolunji Ecological Farm in Pudukottai. He and his friends made a 10-acre barren land into fertile cultivable land in the dry Pudukottai district. He planted 52 varieties of trees in the same waste land extending in 20 acres. His organization 'Kudumbam' preserves and regenerates hundreds of native flora and fauna, in order to ensure a sustainable livelihood.

TN_GOVT_XII_Bio_Zoology_EM_chapter 12.indd 212 20-03-2020 10.30.25 AM